ECE 340: Semiconductor Devices

Lecture 10: Drift

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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Drift in a Semiconductor

Assume uniform distribution of carriers: no diffusion

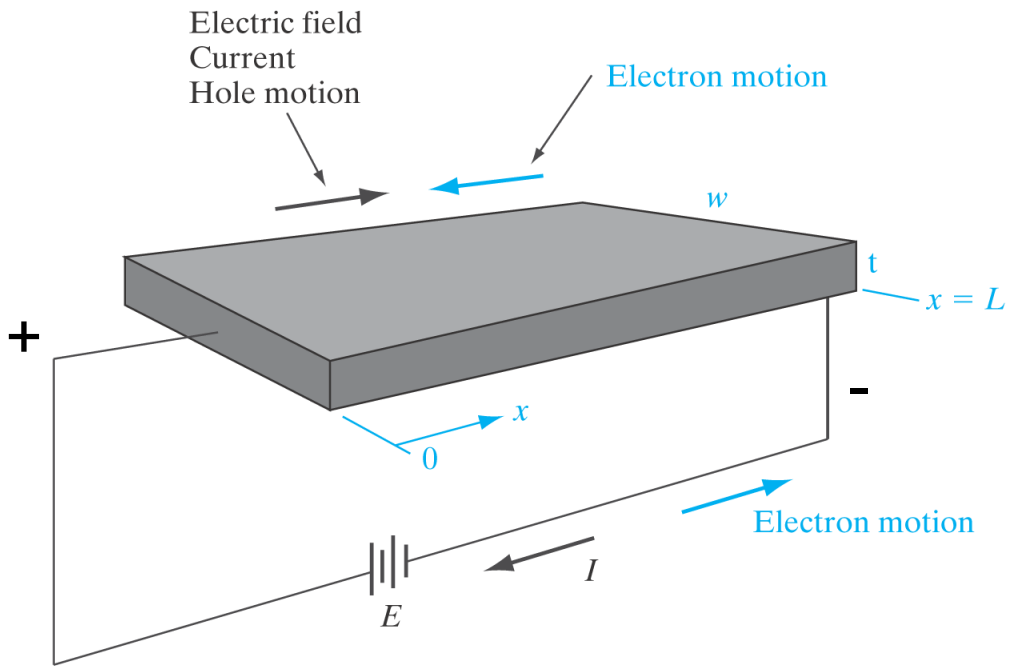


Figure 3.21

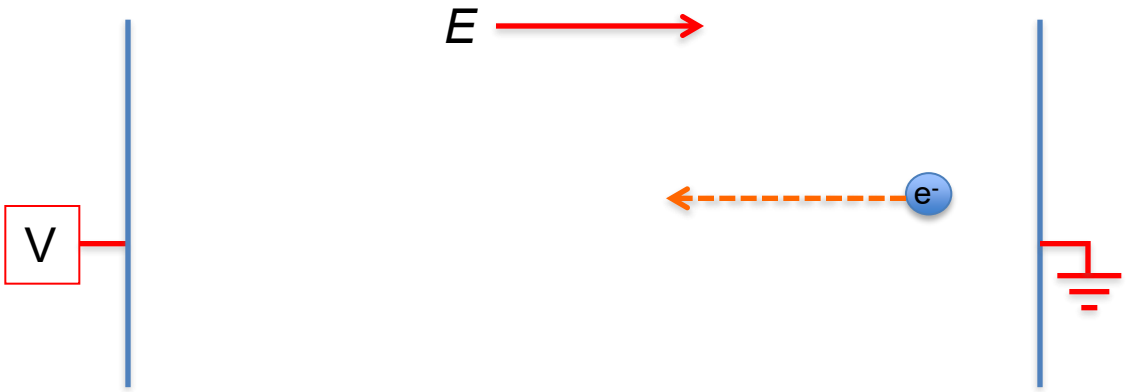
Drift of electrons and holes in a semiconductor bar.

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Electron Motion in a Crystal

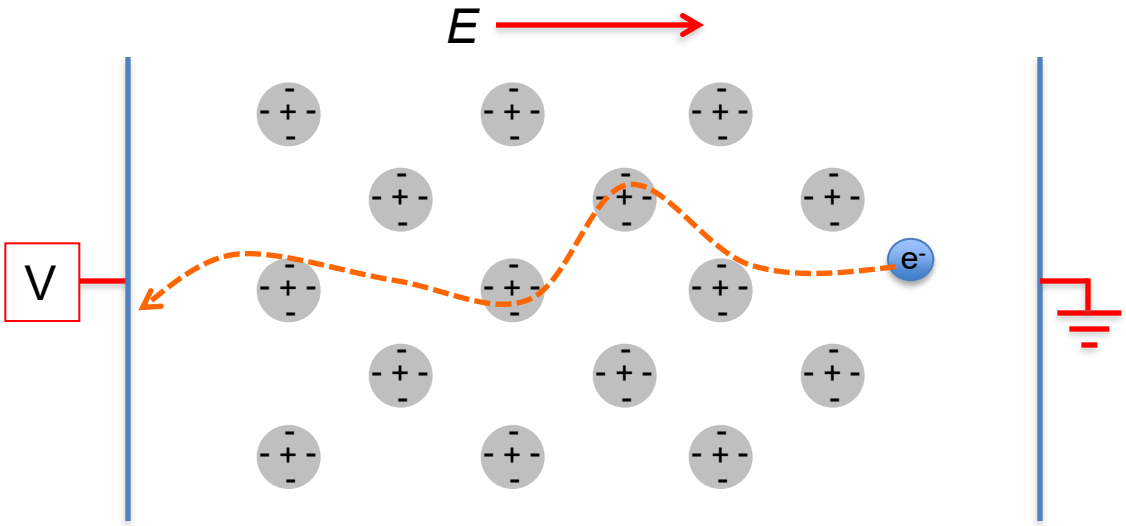
Case 1: in vacuum

- An electron in a vacuum will continue to accelerate under the influence of an electric field



Case 2: In crystal lattice

- An electron traveling through the crystal will experience the periodic potential of the lattice atoms
- The electron will be 'scattered'
- This will impart a momentum change to the electron
- A faster electron will scatter more quickly



Electron Drift in a Constant Electric Field

Force Due to Electric Field:

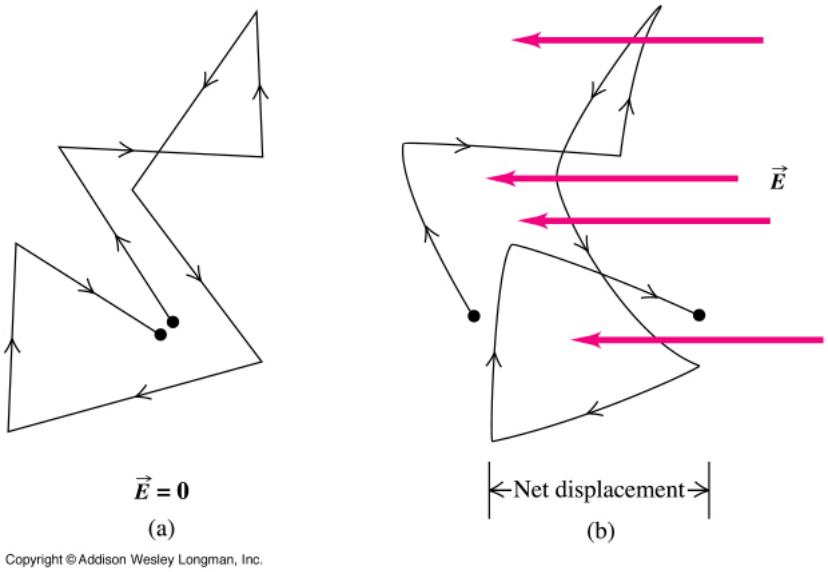
- Electric field creates a net force on each electron: **F = -qE**

$$\text{Force for "n" electrons} = (-q)nE_x = ma = m\left.\frac{dv_x}{dt}\right|_{field} = \left.\frac{dp_x}{dt}\right|_{field}$$

- This force creates a **net momentum** for the population of electrons in the direction of the field

There Must Be an Offsetting Force:

- Scattering events (electron collisions) limit the velocity of the electron
- The net rate of change of momentum, including collisions, is zero under steady-state conditions (otherwise, current would not be constant).



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Force for a single electron:

$$F = -qE_x$$

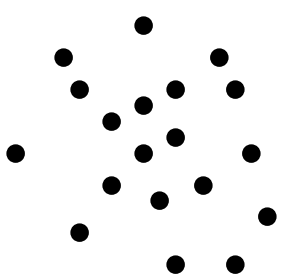
Force for "n" electrons:

$$-nqE_x = \left.\frac{dp_x}{dt}\right|_{field}$$

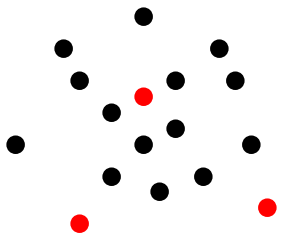
Mean Free Time and Collision Probability

- We start with an initial population of N_0 electrons (or holes) at time t=0
- We assume that any given electron has a fixed probability in time P of having a collision (scattering)
- If we think of scattering centers as fixed points, an electron with average velocity <v> travels an average distance <l> before colliding
- The average time between collisions is thus:

$$\langle t \rangle = \frac{\langle l \rangle}{\langle v \rangle} \equiv \bar{t}$$



Some time later "t"



N(t): # of electrons not collided.

- The probability in time Δt of having a collision is therefore given by:

$$P = \frac{\Delta t}{\bar{t}}$$

• Experienced Collision

Change in Carrier Population

If the probability of a single collision in an interval Δt is given by:

$$P = \frac{\Delta t}{\overline{t}}$$

The incremental number of electrons colliding

in an interval Δt is given by:

$$N(t) - N(t + \Delta t) = PN(t) = \frac{\Delta t}{\overline{t}} N(t)$$

Rewriting this:

$$\frac{N(t) - N(t + \Delta t)}{\Delta t} = \frac{1}{\overline{t}} N(t)$$

Taking the limit as $\Delta t \to 0$:

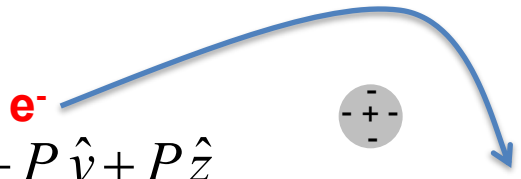
$$\lim_{\Delta t \to 0} \left(\frac{N(t) - N(t + \Delta t)}{\Delta t} \right) = - \lim_{\Delta t \to 0} \left(\frac{N(t + \Delta t) - N(t)}{\Delta t} \right) = - \frac{dN(t)}{dt} = \frac{1}{\overline{t}} N(t)$$

Momentum Transfer from Collisions

From the Previous Slide:

$$\frac{dN(t)}{dt} = -\frac{1}{\bar{t}}N(t) \text{ so:}$$

$$N(t) = N_0 e^{-t/\bar{t}}$$



$$\vec{P} = P_x\hat{x} + P_y\hat{y} + P_z\hat{z}$$

$$\vec{P} = \left(P_x - \langle p_x \rangle\right)\hat{x} + P_y\hat{y} + P_z\hat{z}$$

If the momentum change (loss) due to collisions for the entire population is $-p_x$ the incremental momentum change dp_x in a time interval dt is given by:

$$\left.\frac{dp_x}{dt}\right|_{collisions} = -\frac{p_x}{\bar{t}}$$

p_x: total momentum change if all electrons have collided (and randomized the directed momentum)

Note that the average momentum change per electron is given by: $\langle p_x \rangle = \frac{p_x}{n}$

Steady State Condition

As electrons (holes) go faster, the mean free time between collisions decreases and as a consequence the momentum transfer to the electrons (holes) occurs at a faster rate.

Faster momentum transfer equates to higher force.

Electrons (holes) therefore accelerate until the force due to momentum transfer from scattering is equal to the opposite of the applied force due to the electric field. This is the condition under steady state:

$$\left.\frac{dp_x}{dt}\right|_{field} = -\left.\frac{dp_x}{dt}\right|_{collisions}$$ so, for electrons: (recall F = dp/dt)

$$-nqE_x = \frac{p_x}{\bar{t}}$$ rewriting $$\frac{p_x}{n} = \langle p_x \rangle_{\text{per electron}} = -q\bar{t}E_x = m_n^* \langle v_x \rangle$$

and so: $$\langle v_x \rangle = -\frac{q\bar{t}}{m_n^*}E_x$$ (steady-state velocity)

Summary

- Consider N_o electrons, and the case where the probability of a collision at any time is a constant. If N(t) is the number of electrons that have not undergone a collision at time t, then: $-\frac{dN(t)}{dt} = \frac{1}{\bar{t}_n} N(t)$
- This has a solution:

  $N(t) = N_o e^{-t/\bar{t}_n}$ and $\bar{t}_n$ is the mean free time between scattering events
- The probability of an electron collision in an interval of time dt is $P = \frac{dt}{\bar{t}_n}$
- If a collision imparts a momentum change $-p_x$, the incremental momentum change in an increment of time dp_x is $dp_x = -p_x \frac{dt}{\bar{t}_n}$ so $\left.\frac{dp_x}{dt}\right|_{collision} = -\frac{p_x}{\bar{t}_n}$
- As electron velocity increases, the mean free time between collisions decreases. The force due to the electric field thus becomes balanced by the opposing force due to collisions:

$$F_{collision} + F_{field} = \left.\frac{dp_x}{dt}\right|_{collision} + \left.\frac{dp_x}{dt}\right|_{field} = -\frac{p_x}{\bar{t}_n} - qnE_x = 0$$

- The average momentum imparted to each electron is:

$$\frac{p_x}{n} = \langle p_x \rangle = -q\bar{t}_n E_x \text{ so } \langle v_x \rangle = \frac{\langle p_x \rangle}{m_n^*} = -\frac{q\bar{t}_n}{m_n^*} E_x$$

Comments: Mobility and Field

$$\langle v_x \rangle = \frac{\langle p_x \rangle}{m_n^*} = -\frac{q\bar{t}_n}{m_n^*} E_x$$

$$\mu = \left| \frac{\bar{t}\, q}{m^*} \right|$$

$$\langle v \rangle = \mp \mu E \quad (\text{- electron}, \text{+ hole})$$

- The mean-time between collisions is a property of the temperature in the low-field condition

Electron Current in a Constant Field

$$J_x = -qn\langle v_x \rangle = qn\frac{q\bar{t}_n}{m_n^*}E_x = \frac{nq^2\bar{t}_n}{m_n^*}E_x = \sigma_n E_x = qn\mu_n E_x$$

$$\frac{\text{ampere}}{\text{cm}^2} = \frac{\text{coulomb}}{\text{electron}} \bullet \frac{\text{electrons}}{\text{cm}^3} \bullet \frac{\text{cm}}{\text{s}}$$

$$\frac{\text{C}}{\text{cm}^2\,\text{s}} = \frac{\text{A}}{\text{cm}^2}$$

Electron Conductivity: $\sigma_n = qn\mu_n$ $\left(\text{units: } \frac{C}{V \cdot s \cdot cm} = S/cm\right)$

Electron Mobility: $\mu_n = \frac{q\bar{t}_n}{m_n^*}$ $\left(\text{units: } \frac{cm^2}{V \cdot s}\right)$

Hole Current in a Constant Field

$$J_x = qp\langle v_x \rangle = qp\frac{q\bar{t}_p}{m_p^*}E_x = \frac{pq^2\bar{t}_p}{m_p^*}E_x = \sigma_p E_x = qp\mu_p E_x$$

$$\frac{\text{ampere}}{\text{cm}^2} = \frac{\text{coulomb}}{\text{hole}} \bullet \frac{\text{holes}}{\text{cm}^3} \bullet \frac{\text{cm}}{\text{s}}$$

Hole Conductivity: $\sigma_p = qp\mu_p$

Hole Mobility: $\mu_p \square \frac{q\bar{t}_p}{m_p^*}$

Mobility Comments

- Examining the previous analysis, it can be seen that:

$$\mu_n = -\frac{\langle v_x \rangle}{E_x} \quad \text{and} \quad \mu_p = \frac{\langle v_x \rangle}{E_x}$$

- Mobility is thus a measure of the magnitude of the average velocity of the carrier per unit of electric field
- Because mobility is related to scattering, it will be found that this is a complex topic. There are many scattering mechanisms with different temperature dependences.

Total Current Density

$$J_{x,n} = -qn\langle v_x \rangle = qn\frac{q\bar{t}_n}{m_n^*}E_x = \frac{nq^2\bar{t}_n}{m_n^*}E_x = \sigma_n E_x = qn\mu_n E_x$$

$$J_{x,p} = qp\langle v_x \rangle = qp\frac{q\bar{t}_p}{m_p^*}E_x = \frac{pq^2\bar{t}_p}{m_p^*}E_x = \sigma_p E_x = qp\mu_p E_x$$

Therefore:

$$J_{x,total} = q(n\mu_n + p\mu_p)E_x = \sigma E_x$$

Electron and Hole Current in a Constant Field

Electron Current Relationships

$$J_{x,n} = -qn\langle v_x \rangle = qn\mu_n E_x = \sigma_n E_x$$

Electron Conductivity: $\sigma_n = qn\mu_n$

Electron Mobility: $\mu_n = \frac{q\bar{t}_n}{m_n^*}$

Hole Current Relationships

$$J_{x,p} = qp\langle v_x \rangle = qp\mu_p E_x = \sigma_p E_x$$

Hole Conductivity: $\sigma_p = qp\mu_p$

Hole Mobility: $\mu_p = \frac{q\bar{t}_p}{m_p^*}$

Combined Current

$$J_{x,total} = q(n\mu_n + p\mu_p)E_x = \sigma E_x$$

$$\sigma = q(n\mu_n + p\mu_p)$$

$V = IR$, or $I = \frac{1}{R}V$

We've made it to V=IR!

Example: Intrinsic Silicon

Assume uniform distribution of carriers: no diffusion

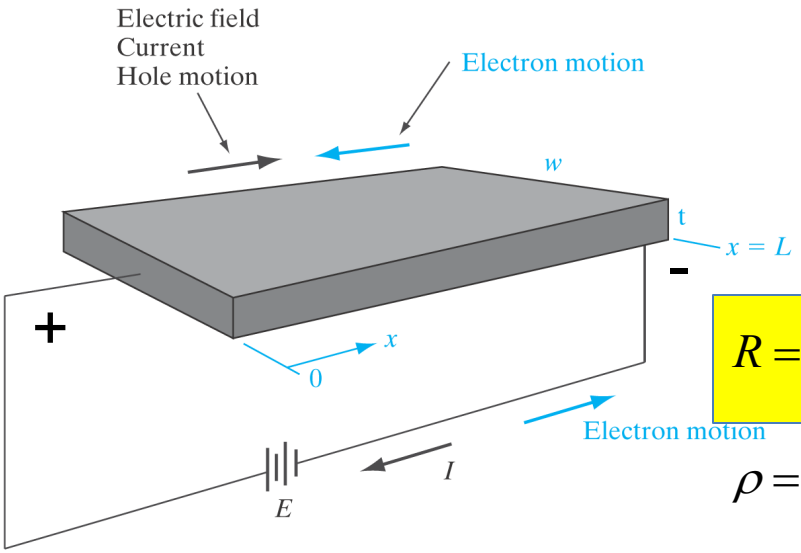


Figure 3.21

Drift of electrons and holes in a semiconductor bar.

$$J_n = -qn\langle v_n \rangle = -qn(-\mu_n E_x)$$

$$J_p = qp\langle v_p \rangle = qp(\mu_p E_x)$$

$$R = \frac{\rho L}{A} = \frac{\rho L}{wt} = \frac{L}{wt}\frac{1}{\sigma}$$

$$\rho = \frac{1}{\sigma}$$

$$\rho(\text{Intrinsic Silicon}) = \frac{1}{\sigma} = \frac{1}{q(n\mu_n + p\mu_p)}$$

$$= \frac{1}{1.6x10^{-19}(1.5\times10^{10}\,\square 1350 + 1.5\times10^{10}\,\square 480)}$$

$$= \frac{1}{4.39\times10^{-6}(\Omega - cm)^{-1}} = 2.28\times10^{5}\,\Omega - cm$$

We know how to calculate n and p

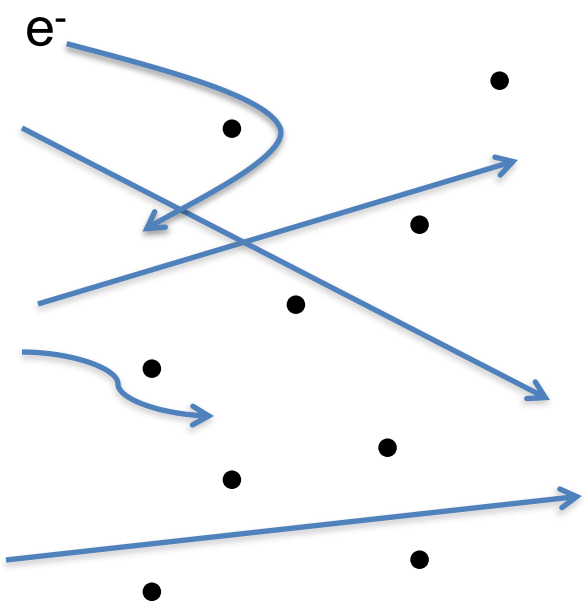
What about μ?

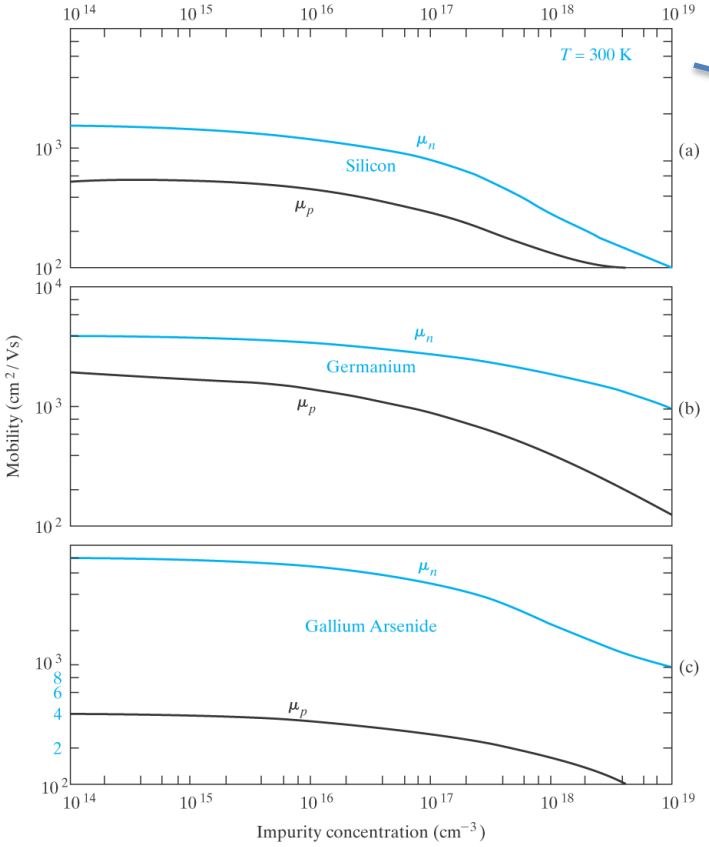
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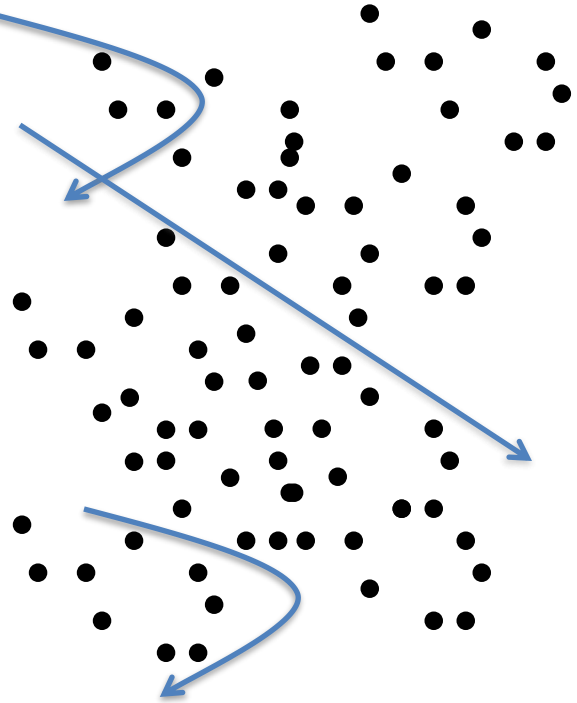
Effects of Impurity Concentration and Temperature on Mobility

Mobility Impurity Dependence

More scattering centers translates to reduced mobility







$$\mu = \bar{t}\,\frac{q}{m^*}$$

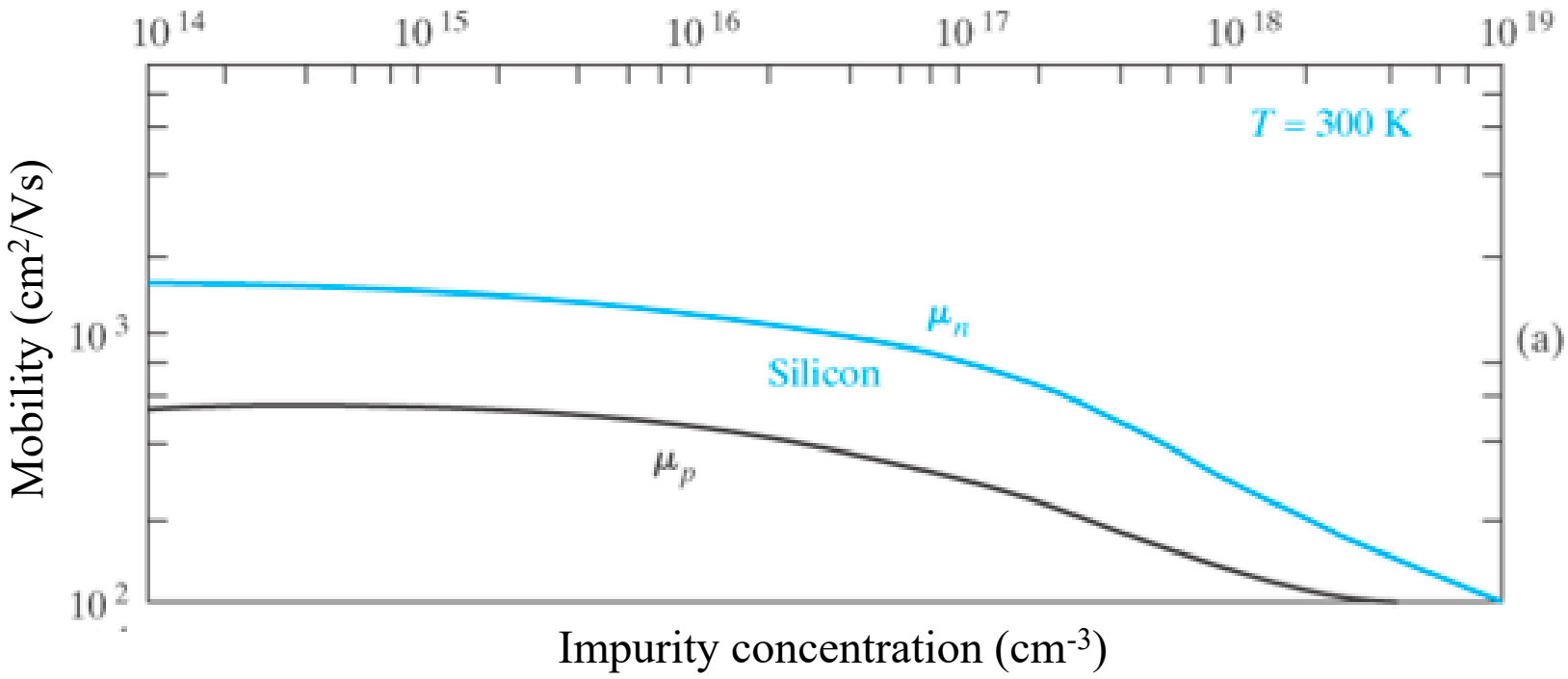
Figure 3.23

Variation of mobility with total doping impurity concentration ($N_a + N_d$) for Ge, Si, and GaAs at 300 K.

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What Value Do I Use for Mobility?

$$\mu \sim \frac{q\bar{t}}{m^*}$$



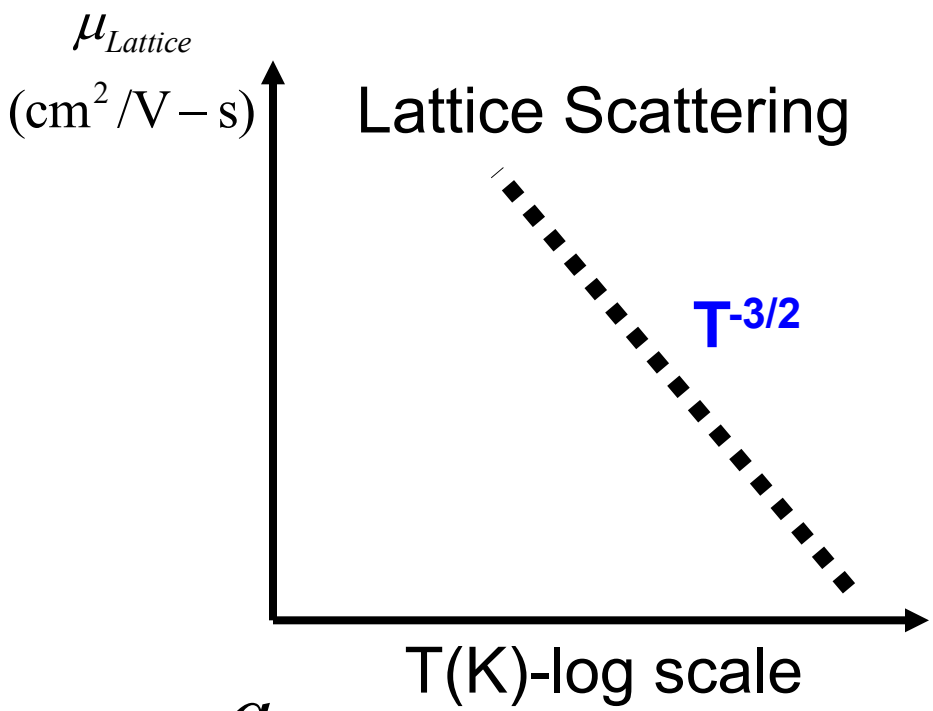
Impurity Concentration, Not Carrier Concentration

N_D+N_A

Temperature Effect on Lattice Scattering

- Electron and hole mobility is a function of the frequency of collision events (scattering)
- **Lattice** Scattering
  - A carrier moving through the crystal is scattered by a vibration of the lattice (phonon)
  - At temperature increases, there is more vibration of the lattice
  - The mobility decreases as the temperature increases due to greater thermal agitation of the lattice
  - At higher temperature, the mean time between collisions is also smaller

$$\mu_{Lattice} \propto T^{-3/2}$$

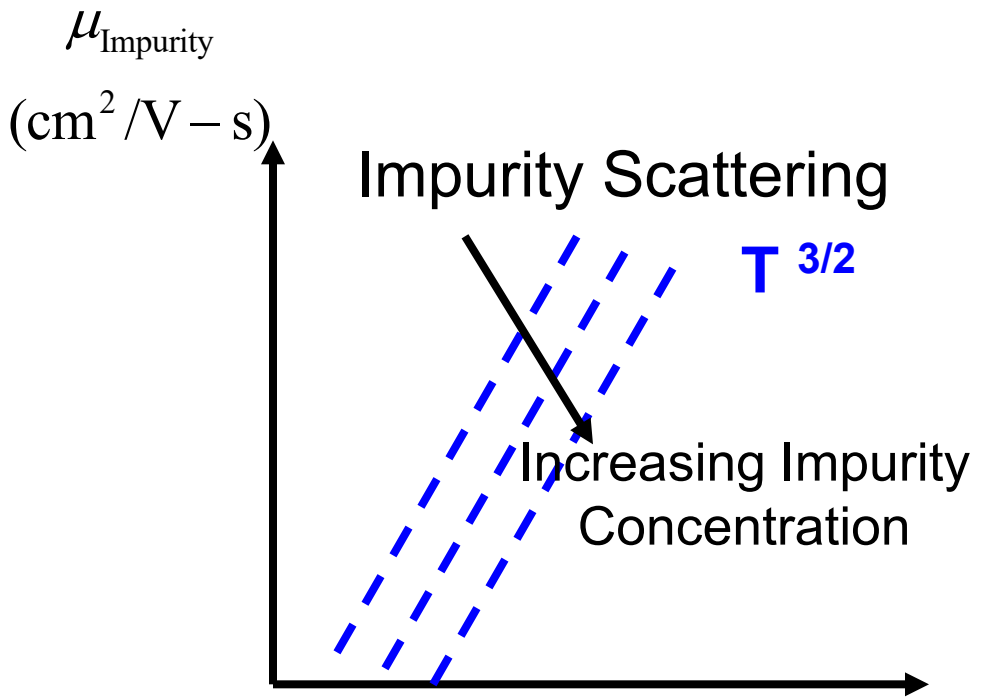


$$\mu = \bar{t}\,\frac{q}{m^*}$$

Temperature Effect on Impurity Scattering

- **Impurity** Scattering
  - Ionized impurity scattering refers to the interaction with charged ions (carrier freeze out not considered)
  - A carrier moving through the crystal is scattered by ionized impurities
  - The **mobility decreases** as the **impurity concentration increases** since there are more scattering centers
  - The **mobility decreases** as the **temperature decreases** since the carrier itself has less thermal energy and will be scattered more strongly by the impurity

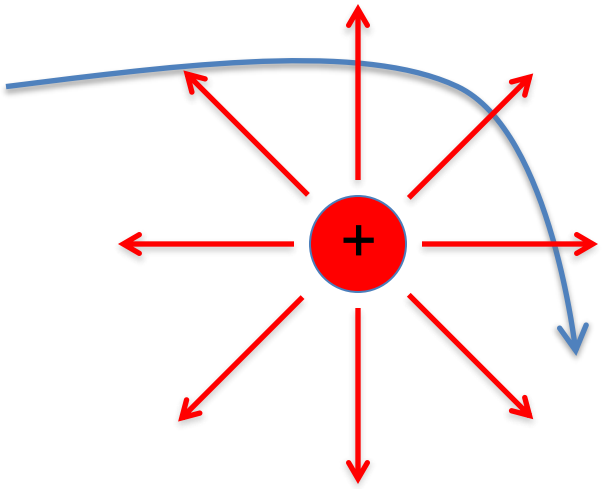
$$\mu_{\text{Impurity}} \propto T^{3/2}$$



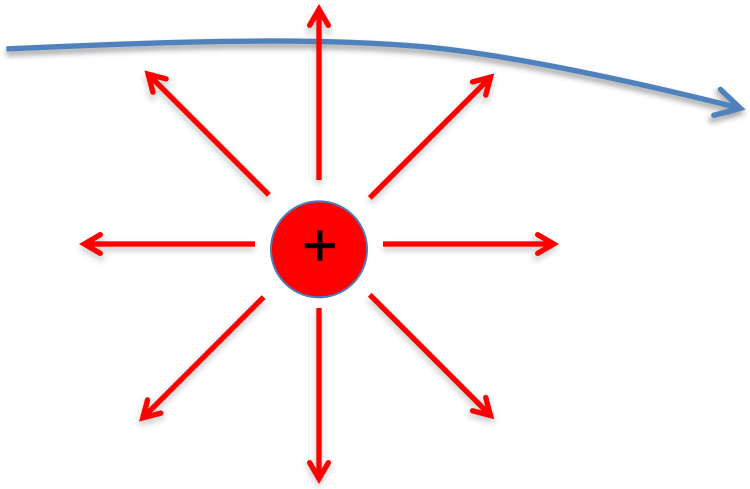
$$\mu = \overline{t}\,\frac{q}{m^*}$$

Fast Electron Versus Slow Electron

Slow Electron
(Cold Temperature)



Fast Electron
(Warm Temperature)



- The slow electron interacts with the fixed field for a longer period of time, deflecting it from the direction of propagation

Combined Effects

Mobilities Add Inversely:

$$\frac{1}{\mu_{tot}} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_3} + \ldots$$

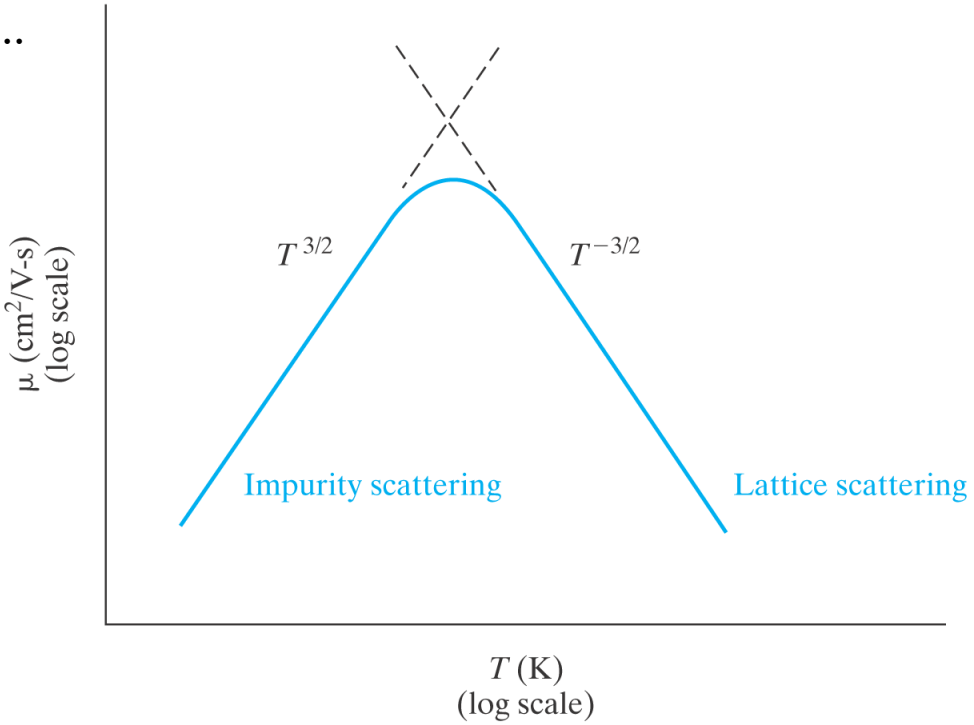


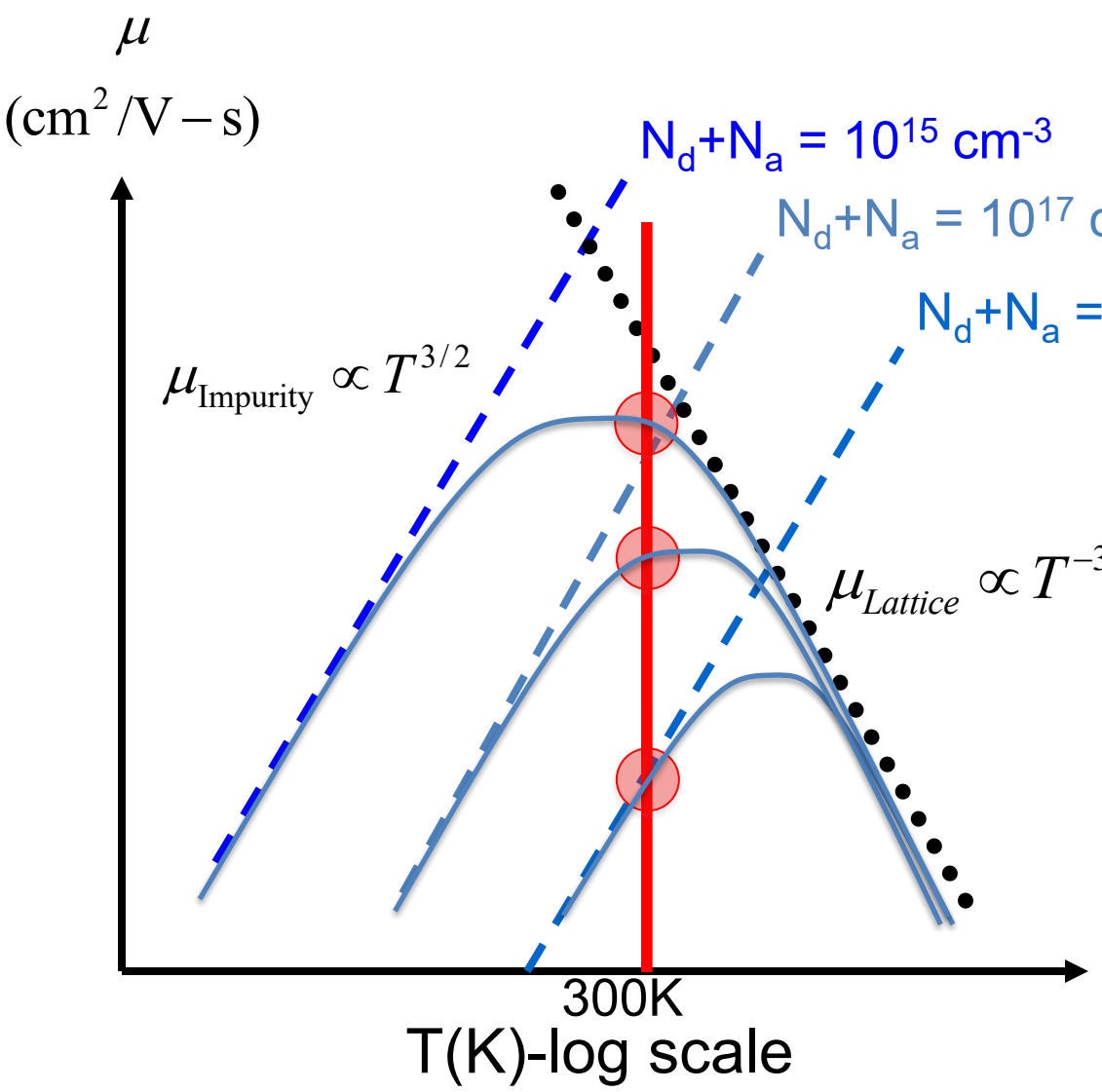
Figure 3.22

Approximate temperature dependence of mobility with both lattice and impurity scattering.

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Effects of Both Temperature and Impurity Concentration on Mobility

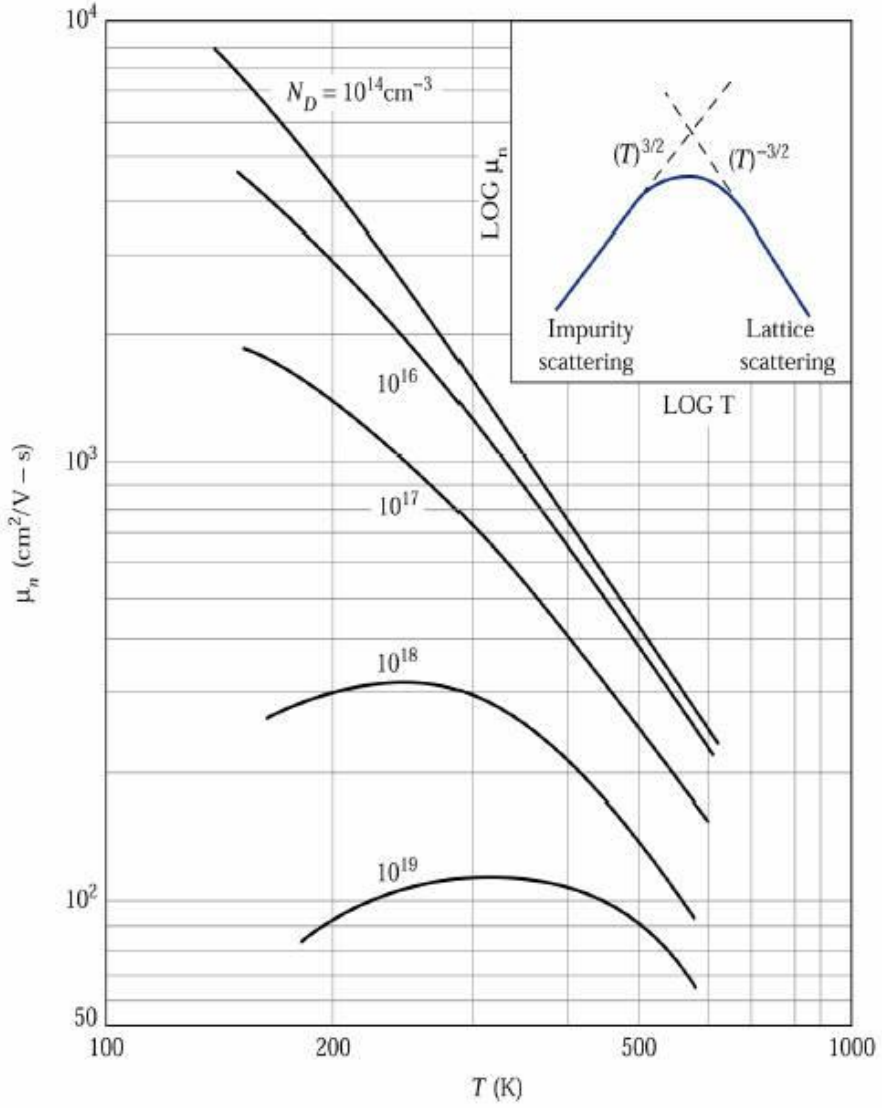
$$\frac{1}{\mu} = \frac{1}{\mu_{Impurity}} + \frac{1}{\mu_{Lattice}} + \frac{1}{\mu_{Defect}} + \ldots$$



At T=300K

- For $N_d+N_a=10^{19}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{17}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{15}$cm^{-3}, the mobility is dominated by lattice scattering

Comparison With Measured Data



http://www.globalsino.com/micro/1/1micro9936.html